Solutions for Graduate Combinatorics HW

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Due to limited time the following are only **sketches** of full solutions, and in particular these solutions alone wouldn't necessarily constitute a solution worth full marks. I also emphasize that there may exist other (possibly simpler) solutions these problems.

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1 HW1

1.1 *Let $\mathcal{F}$ be an r-uniform set system with the property that $F' \cap F'' \not\subseteq F$ for any distinct $F, F', F'' \in \mathcal{F}$. Prove that $|\mathcal{F}| \leq 1 + \binom{r}{\lfloor r/2 \rfloor}$ (Hint: let $F'' \in \mathcal{F}$ be arbitrary and consider the set system $\{F \cap F'' : F \in \mathcal{F} \setminus \{F''\}\}$) [2-].*

The set system $\mathcal{F}''$ as suggested in the hint is an antichain: for distinct $F, F' \in \mathcal{F}$, having $F' \cap F'' \subseteq F \cap F''$ would in particular imply $F' \cap F'' \subseteq F$ which we assumed not to be the case. Moreover, this same argument shows $|\mathcal{F}''| = |\mathcal{F} \setminus \{F''\}|$, i.e. that $F \cap F'' \neq F' \cap F''$ for distinct F, F'. Since $\mathcal{F}''$ has ground set F'' of size r, it follows from Sperner's Theorem

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that $|\mathcal{F}''| \leq \binom{r}{\lfloor r/2 \rfloor}$, and hence

$$|\mathcal{F}| = 1 + |\mathcal{F} \setminus \{F''\}| \leq 1 + \binom{r}{\lfloor r/2 \rfloor}.$$

1.5 *Prove* **??** *(Hint: try proving the weaker lower bound of n^{-2^d} to get a feel for the proof idea) [2].*

The case $d = 1$ is Dilworth's Theorem. Inductively assume we have proven the result up to some value $d \geq 2$. Let G Be a graph and $P_1, \ldots, P_d$ posets on $V(G)$ such that $x \sim y$ iff $x <_i y$ or $y <_i x$ for some i. If P_d contains a chain on at least $n^{1/(d+1)}$ elements then this corresponds to a clique of size at least $n^{1/(d+1)}$ in G. By the Corollary of Dilwrorth's Theorem that $n \leq h(P)w(P)$ for all P, we can assume P_d contains an antichain on at least $n^{d/(d+1)}$ vertices of G. Let $G' \subseteq G$ be the induced subgraph of G on this antichain. Since G' is induced on an antichain of P_d, it follows that $P_1, \ldots, P_{d-1}$ are such that two vertices of G' are adjacent if and only if they are comparable by some P_i with $1 \leq i \leq d-1$, so inductively G' has either a clique or independent set of size at least $(n^{d/(d+1)})^{1/d} = n^{1/(d+1)}$, proving the result.

1.7a *Prove Erdős-Szekeres using Dilworth's Theorem [2].*

Construct a poset P on $[n]$ where $i <_P j$ if and only if $a_i < a_j$ and $i < j$ (with these latter inequalities being with respect to the usual ordering on $\mathbb{R}$). By Dilworth there exists a chain or an antichain of size at least $\sqrt{n}$ of P, and it is straightforward to check that these correspond to increasing or decreasing subsequences of π of the same size.

1.7b *Prove this without using Dilworth's Theorem (Hint: for each j define i_j to be the length of the longest increasing subsequence which ends at a_j and d_j to be the length of the longest decreasing subsequence which ends at a_j. Can we have $(i_j, d_j) = (i_{j'}, d_{j'})$ for $j < j'$?) [2+].*

Following the hint, we can not have $(i_j, d_j) = (i_{j'}, d_{j'})$ for $j < j'$: if $a_j < a_{j'}$ then we could add $a_{j'}$ to the end of the increasing subsequence of length i_j ending at a_j to find $i_{j'} > i_j$; an analogous argument works if $a_j > a_{j'}$. It follows that the n pairs (i_j, d_j) are distinct, which means we can not have $i_j, d_j < \sqrt{n}$ for all n and hence some i_j or d_j is at least this large, corresponding to a large increasing or decreasing subsequence.

1.8 *Prove Mirsky's Theorem: if $P = (X, \leq)$ is a finite poset of height $h(P)$, then there exists a partition of X into $h(P)$ antichains (Hint: prove that the set of maximal elements M of P form an antichain) [2].*

For the hint, if two maximal elements x, y were comparable, say with $x < y$, then x would not be maximal. We conclude that M is indeed an antichain.

We now prove the main result by induction on $h(P)$, the case $h(P) = 0$ being trivial since this can only occur if $P = \emptyset$. We claim that $P - M$ has height at most $h(P) - 1$; indeed, if $P - M$ had a chain of length $h(P)$ then whatever element is at the top of the chain is not maximal and hence could be extended to a chain of length $h(P) + 1$ in P, a contradiction. Inductively then we can decompose $P - M$ into at most $h(P) - 1$ antichains, which together with M gives the desired partition.

2 HW2

1.2b *Prove for any integer $t \geq 1$ that*

$$\sup_{r\in\mathbb{R}} \Pr[\sum_i a_i X_i \in [r, r+2t)] \leq \max_{T\subseteq\binom{\{0,1,\ldots,n\}}{t}} \sum_{r\in t} \binom{n}{r} 2^{-n},$$

where to be clear this upper bound is equal to 2^{-n} times the sum of the t largest binomial coefficients [2].

As in the Littlewood-Offord proof, we may assume $a_i > 0$ for all i. Fix an arbitrary r and let $\mathcal{F}$ be the set of $F \subseteq [n]$ with the property that $\sum_{i\in F} a_i - \sum_{i\notin F} a_i \in [r, r+t+1)$. By Erdős's Theorem it suffices to prove that $\mathcal{F}$ does not have a chain of size larger than t. Say for contradiction that it contained some $F_1 \subsetneq \cdots \subsetneq F_{t+1}$. Define $\sigma_j = \sum_{i\in F_j} a_i - \sum_{i\notin F_j} a_i$. It is straightforward to prove by induction that $\sigma_j \geq 2(j-1) + \sigma_1$ since each a_i is at least 1. In particular since $\sigma_1 \geq r$ by hypothesis of $F_1 \in \mathcal{F}$ we find $\sigma_{t+1} \geq 2t + r$, a contradiction to $F_{t+1} \in \mathcal{F}$ implying $\sigma_{t+1} < 2t + r$.

2.1 *We consider an abstract version of the Katona circle argument: for a permutation $\pi : [n] \to [n]$, we define for $F = \{x_1, \ldots, x_r\} \subseteq [n]$ the set $\pi(F) = \{\pi(x_1), \ldots, \pi(x_r)\}$ and for $\mathcal{F} \subseteq 2^{[n]}$ we define $\pi(\mathcal{F}) = \{\pi(F) : F \in \mathcal{F}\}$. Prove that for any two families $\mathcal{F}, \mathcal{A} \subseteq 2^{[n]}$ that*

$$\sum_{F\in\mathcal{F}} \frac{\left|\mathcal{A} \cap \binom{[n]}{|F|}\right|}{\binom{n}{|F|}} \leq \max_\pi |\pi(\mathcal{F}) \cap \mathcal{A}|,$$

where the maximum ranges over all permutations $\pi : [n] \to [n]$ [2-].

We count all pairs (π, F) with $\pi(F) \in \mathcal{A}$ in two ways. Each of the $n!$ bijections π is trivially in at most $\max_\pi |\pi(\mathcal{F}) \cap \mathcal{A}|$ pairs. On the other hand, for each F we can identify (π, F) by specifying the image $\pi(F) \in \mathcal{A} \cap \binom{[n]}{|F|}$ and given this we have $|F|!(n-|F|)! = \frac{n!}{\binom{n}{|F|}}$ ways to specify π. Setting these two counts equal to each other and dividing both sides by $n!$ gives the stated bound.

2.2 *Give an alternative proof of the LYM inequality by using that if $\mathcal{I}$ is the set of initial intervals $\{1, 2, \ldots, i\}$ for $1 \leq i \leq n$ then every antichain of $\mathcal{I}$ has size at most 1 [2-].*

This follows from the previous exercise applied with $\mathcal{A} = \mathcal{I}$ after observing that if $\mathcal{F}$ is an antichain, then at most one $F \in \mathcal{F}$ can have $\pi(F)$ an initial interval (since a second F' with, say, $\pi(F')$ an initial interval containing the initial interval $\pi(F)$ would imply $F \subseteq F'$).

2.5 *We say that a family $\mathcal{F} \subseteq 2^{[n]}$ is union if $F \cup F' \neq [n]$ for all $F, F' \in \mathcal{F}$. Prove that $\mathcal{F}$ is a union family if and only if*

$$\overline{\mathcal{F}} := \{[n] \setminus F : F \in \mathcal{F}\}$$

is intersecting. In particular, any problem about intersecting families can be turned into a problem about union families, which in practice can sometimes be easier to solve [1+].

Any $x \in F \cap F'$ satisfies $x \notin ([n] \setminus F) \cup ([n] \setminus F') = [n] \setminus (F \cap F')$, giving the result.

3 HW3

2.6a *Prove that every maximal intersecting family $\mathcal{F} \subseteq 2^{[n]}$ has $|\mathcal{F}| = 2^{n-1}$. For this we emphasize that by definition, no intersecting family $\mathcal{F}$ can have $\emptyset \in \mathcal{F}$ since then $F \cap F' = \emptyset$ for $F = F' = \emptyset$ elements in $\mathcal{F}$ [2].*

Let $\mathcal{F}$ be maximal intersecting. We prove the stronger fact that for every $F \subseteq [n]$ exactly 1 of F or $[n] \setminus F$ is in $\mathcal{F}$. Certainly at most 1 of these sets can be in $\mathcal{F}$ if it is intersecting, so now assume for contradiction that there is some F such that neither is in $\mathcal{F}$. If $\{F\} \cup \mathcal{F}$ is intersecting then this contradicts $\mathcal{F}$ being maximal, so there must be some $F' \in \mathcal{F}$ disjoint from F. This means $F' \subseteq [n] \setminus F$, and this in turn means that $\{[n] \setminus F\} \cup \mathcal{F}$ is intersecting since any F'' disjoint from this would be disjoint with F'.

2.6b *Prove that there exist maximal intersecting families $\mathcal{F} \subseteq \binom{[n]}{r}$ with $|\mathcal{F}| \leq 4^r$ for all n, r [2].*

We trivially have $|\mathcal{F}| \leq 2^n$ for any family $\mathcal{F}$ so it suffices to prove the result when $n \geq 2r$. There are many possible solutions, the simplest is to take $\mathcal{F} = \binom{[2r-1]}{r}$ which is intersecting but any $F \not\subseteq [2r-1]$ will necessarily intersect $[2r-1]$ in at most $r-1$ vertices and hence be disjoint from some edge in $\mathcal{F}$, so $\mathcal{F}$ is maximal with size at most $2^{2r-1} \leq 4^r$.

2.8 *Recall that for a family $\mathcal{F} \subseteq 2^{[n]}$ we define its Lubell measure*

$$\lambda(\mathcal{F}) = \sum_{F \in \mathcal{F}} \frac{1}{\binom{n}{|F|}}.$$

Prove that if $\mathcal{F}$ is intersecting then $\lambda(\mathcal{F}) \leq \frac{n+1}{2}$ [2].

Proof without circle method: The key idea is that for each F, at most one of F and $[n] \setminus F$ is in $\mathcal{F}$. The slick way of doing this is:

$$2\lambda(\mathcal{F}) = \sum_{F \in \mathcal{F}} 2\binom{n}{|F|}^{-1} = \sum_{F \in \mathcal{F}} \binom{n}{|F|}^{-1} + \binom{n}{|[n] \setminus F|}^{-1} \leq \sum_{F \in 2^{[n]}} \binom{n}{|F|}^{-1} = n+1,$$

with the last equality using that each of the $n+1$ layers has Lubell weight 1.

Proof with circle method: Take $\mathcal{A}_n^*$ to be the family of all cyclic intervals except for $[n]$. Any $\mathcal{F} \subseteq \mathcal{A}_n^*$ which is intersecting has size at most $|\mathcal{A}_n^*| = \binom{n}{2}$ again simply because an intersecting family can not contain both a set and its complement. Applying the abstract Katona circle argument from Exercise 2.1 then yields that any intersecting $\mathcal{F} \subseteq 2^{[n]} \setminus [n]$ has $\lambda(\mathcal{F}) = \sum_F \binom{n}{|F|}^{-1} \leq \frac{n-1}{2}$, and this in turn implies that any intersecting $\mathcal{F} \subseteq 2^{[n]}$ has $\lambda(\mathcal{F}) \leq 1 + \frac{n-1}{2}$ since $\lambda([n]) = 1$.

We emphasize that a direct argument with $\mathcal{A}_n$ instead of $\mathcal{A}_n^*$ does not quite work out because $|\mathcal{A}_n \cap \binom{[n]}{n}$ does not equal n like in all the other terms but rather 1.

4 HW4

3.8 *Prove that for every family $\mathcal{F}_1 \subseteq 2^{[n]}$ there is some finite sequence of integers $i_1, \ldots, i_t$ such that if we define $\mathcal{F}_{k+1} = s_{-i_k}(\mathcal{F}_k)$ then $\mathcal{F}_{t+1}$ is a downset [2-].*

Similar to our proof of **??** we take $\mathcal{F}_{k+1} = s_{-i_k}(\mathcal{F}_k)$ whenever there exists i_k with $\mathcal{F}_k \neq s_{-i_k}(\mathcal{F}_k)$. One can easily check then that the quantity

$$\sum_{e \in \mathcal{F}_k} |e|$$

strictly decreases in k, so at some point this process must stop at some $\mathcal{F}_{t+1}$ which has $\mathcal{F}_{t+1} = s_{-i}(\mathcal{F}_{t+1})$ for all i, and it is straightforward to check that this implies $\mathcal{F}_{t+1}$ is a downset.

3.10 *We say a family $\mathcal{F}$ has diameter d if $\max_{F,F' \in \mathcal{F}} |F \Delta F'| = d$ where here $F \Delta F'$ denotes the symmetric difference, and we let $\mathrm{diam}(\mathcal{F})$ denote the diameter of $\mathcal{F}$. Prove that $\mathrm{diam}(s_{-i}(\mathcal{F})) \leq \mathrm{diam}(\mathcal{F})$ for all families $\mathcal{F}$ and integers i [2-].*

Let $F, F' \in \mathcal{F}$ be such that $\mathrm{diam}(s_{-i}(\mathcal{F})) = |s_{-i}(F) \Delta s_{-i}(F')|$. If i is not in this symmetric difference then $|s_{-i}(F) \Delta s_{-i}(F')| \leq |F \Delta F'|$ since the only element which possibly changed in these sets is i, and from this it follows that the diameter of $\mathcal{F}$ is at least that of $s_{-i}(\mathcal{F})$. As such we may assume that, say, $i \in s_{-i}(F)$ and $i \notin s_{-i}(F')$. The fact that $i \in s_{-i}(F)$ implies that $F - i \in \mathcal{F}$. This implies that $|s_{-i}(F) \Delta s_{-i}(F')|$ is equal to either $|F \Delta F'|$ or $|(F - i) \Delta F'|$ depending on if $i \in F'$ or not. In either case this shows $\mathrm{diam}(s_{-i}(\mathcal{F})) = |s_{-i}(F) \Delta s_{-i}(F')| \leq \max\{|F \cap F'|, |(F - i) \cap F'|\} \leq \mathrm{diam}(\mathcal{F})$, proving the result.

3.11a *Prove Katona's Union Theorem using Kleitman's Diameter Theorem [1+].*

If $\mathcal{F}$ is such that $|F \cup F'| \leq d$ then certainly $|F \Delta F'| \leq d$, so the bound for Katona follows from the bound of Kleitman.

3.11b *Prove Kleitman's Diameter Theorem using Katona's Union Theorem [2].*

By 3.8 and 3.10 we can assume $\mathcal{F}$ is a downset. We claim then that $\mathcal{F}$ is $(n-d)$-union. Indeed, if we had $|F \cup F'| > d$ then because $\mathcal{F}$ is a downset we have $F'' := F' \setminus F \in \mathcal{F}$, and this satisfies $|F \Delta F''| = |F \cup F'| > d$, a contradiction. We conclude that $\mathcal{F}$ is $(n-d)$-union, so the bound follows from Katona.

5 HW5

3.6 *Given a graph G, let $\kappa(G)$ denote the total number of cliques of G, i.e. this is the number of K_1's of G plus the number of K_2's of G, and so on. For integers m, n with $m = \binom{k}{2}$ for some integer $k \leq n$, determine the largest possible value of $\kappa(G)$ amongst n-vertex graphs with m edges [2+].*

We claim that the answer is $\sum_{i=1}^{k} \binom{k}{i}$. The lower bound comes from a clique on k vertices. For the upper bound, we claim that any graph G with $\binom{k}{2}$ edges has at most $\binom{k}{i}$ cliques of size i. Indeed, let $\mathcal{F}$ denote the set system on $V(G)$ whose hyperedges are cliques of size i. Then $\partial_2 \mathcal{F} \subseteq E(G)$ and hence $|\partial_2 \mathcal{F}| \leq \binom{k}{2}$, which by the contrapositive of the Lovász version of Kruskal-Katona implies $|\mathcal{F}| \leq \binom{k}{i}$, proving the claim and hence the result.

4.2 *Prove for all $r, p \geq 1$ that there exists an r-uniform set system $\mathcal{F}$ with $|\mathcal{F}| = (p-1)^r$ which does not contain a sunflower with p petals [2-].*

The result is trivial for $p = 1$ so assume $p > 1$ from now on. Let $X_1, \ldots, X_r$ be disjoint sets on $p-1$ elements and let $\mathcal{F}$ be the set system consisting of all sets which use exactly 1 vertex from each X_i. We claim this does not have a sunflower with p petals. Assume otherwise with $S_1, \ldots, S_p$ the edges and K the kernel. We must have $|K| < r$ (else $S_i = S_j$ for all i, j, giving a sunflower with $1 < p$ edges), and in particular K is disjoint from some X_i. By construction each edge S_j must use exactly one vertex of X_i and these two vertices can not equal each other since otherwise the sunflower condition would force this vertex in K, a contradiction to K being disjoint from X_i. However, it is impossible for each S_j to use distinct vertices from X_i since $|X_i| < p$, and this gives the desired contradiction.

4.4a *Prove Deza's Theorem when $\ell = 1$ (Hint: prove that if $\mathcal{F}$ contains a sufficiently large sunflower, then $\mathcal{F}$ must be a sunflower) [2].*

We claim that if $\mathcal{F}$ contains a sunflower with at least $r+1$ petals then $\mathcal{F}$ is a sunflower. Indeed, say $\mathcal{F}$ had a largest sunflower with edges $S_1, \ldots, S_p$ for $p \geq r+1$, and by hypothesis this sunflower must have a kernel of size 1, say with $K = \{v\}$. Assume there was some edge $F \in \mathcal{F}$ not part of this sunflower. If $v \in F$ then F by hypothesis must be disjoint from $S_i \setminus \{v\}$ for all i, meaning F can be added to the sunflower to make a larger one. Thus $v \notin F$, which again by hypothesis means F must intersect each of the disjoint sets $S_i \setminus \{v\}$ which is impossible for $p \geq r+1$.

With this claim in mind, consider any $F = \{v_1, \ldots, v_r\} \in \mathcal{F}$. By a similar argument to the above we observe that the set of edges of $\mathcal{F}$ containing each v_i form a sunflower, so by the claim we may assume that the number of edges containing each v_i is at most r (i.e. there are at most $r-1$ such edges other than F itself). Since every edge contains at least one v_i by hypothesis on $\mathcal{F}$, we find that the total number of edges of $\mathcal{F}$ is at most $1 + r(r-1) = r^2 - r + 1$, contradicting our size condition on $\mathcal{F}$.

6 HW6

5.1 *For all integers $r, \nu \geq 1$ construct an r-uniform $\mathcal{F}$ with $\nu(\mathcal{F}) = \nu$ and $\tau(\mathcal{F}) = r\nu$ [2-].*

Take $K^{(r)}_{r\nu+r-1}$ which has matching number ν and transversal number $r\nu$.

5.2 *Prove that if $\mathcal{F}$ is an intersecting family then $\tau(\mathcal{F}) \leq \min_{F \in \mathcal{F}} |F|$ [1+].*

We observe that any edge $F \in \mathcal{F}$ is a transversal since by assumption every edge of $\mathcal{F}$ intersects F, giving the bound.

5.3 *(Gyárfás) Prove that if $\mathcal{F}$ is a set system where every edge has size at most r, then for every set of vertices S the number of transversals T of $\mathcal{F}$ with $S \subseteq T$ and $|T| = \tau(\mathcal{F})$ is at most $r^{\tau(\mathcal{F})-|S|}$ [2].*

We prove the result by backwards induction on S. The base case of $|S| = \tau(\mathcal{F})$ is trivial. Assume that $|S| < \tau(\mathcal{F})$ and that we have proven the result for all sets larger than S.

Because $|S| < \tau(\mathcal{F})$ there is some $F \in \mathcal{F}$ which does not intersect F. Observe that every transversal which contains S must contain $S \cup \{v\}$ for some $v \in F$. Inductively the number of T containing a given $S \cup \{v\}$ is at most $r^{\tau(\mathcal{F})-|S|-1}$, and summing this over each of the $|F| \leq r$ choices for v gives at most $r^{\tau(\mathcal{F})-|S|}$ choices for T, completing the inductive step.

5.4 *(Erdős-Lovász) Prove that if $\mathcal{F}$ is an r-uniform intersecting family with $\tau(\mathcal{F}) = r$ then $|\mathcal{F}| \leq r^r$ [2].*

By the previous problem applied with $S = \emptyset$ there are at most r^r transversals of $\mathcal{F}$ of size r. Because $\mathcal{F}$ is intersecting and r-uniform, every edge is a transversal of size r, so in total this means $|\mathcal{F}| \leq r^r$.